

Critical Analysis and Comparison of State Machines

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Project Motivation

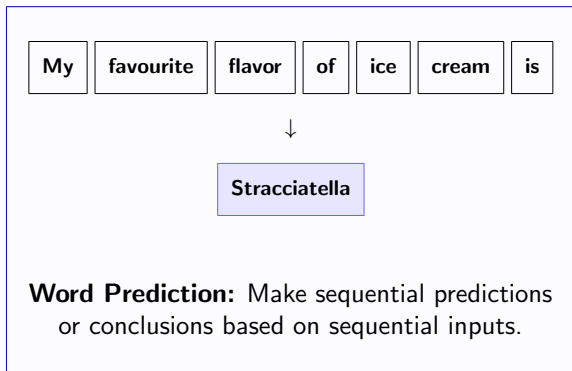
Sequence-to-Sequence Problems:

Map input sequences to output sequences, often with temporal dependencies.

Applications:

- Natural Language Processing
- Speech and Audio Processing
- Code Generation
- Dialog Systems

Example:



Source: [8, pp. 2f., 20–28]

State-of-the-Art Paradigms

1) Recurrent Neural Network:

- Accumulates information in hidden state over time
- Inference: Fast (single hidden state)
- Training: Slow (sequential bottleneck)

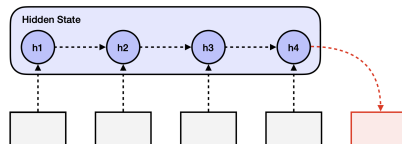


Figure: Overview for prediction of RNN

Source: [2, pp. 370, 372–376], [5]

State-of-the-Art Paradigms

2) Transformer:

- Attention Matrix that sets all elements in relation
- Inference: Slow (attention matrix grows with input sequence)
- Training: Fast (because parallelizable)

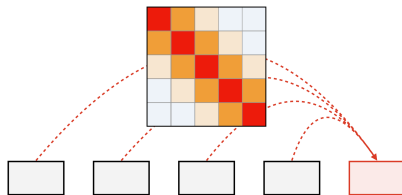


Figure: Overview for prediction of Transformer

Source: [9, pp. 3–8]

State Space Model

1. Continuous Representation

- State Space Representation
- used to model physical systems

2. Recurrent Representation

- Discretization $((A, B, C) \rightarrow (\bar{A}, \bar{B}, \bar{C}))$
- HiPPO initialization of $\bar{A}, \bar{B}, \bar{C}$
(better long-range dependencies)
- SSM is only *linear*
(in comparison to non-linear RNN)

State Space Representation:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t)\end{aligned}\tag{1}$$

SSM:

$$\begin{aligned}x_k &= \bar{A}x_{k-1} + \bar{B}u_k, \\ y_k &= \bar{C}x_k\end{aligned}\tag{2}$$

RNN:

$$h_k = f(h_{k-1}, x_k; \theta)\tag{3}$$

State Space Model: Dual Computation Modes

Training:

- Linearity key for efficient training
- Inputs can be convoluted into single calculation

⇒ **Parallelizable during Training**

Inference:

- Information accumulated in single state
- Inference of next element y_{k+1} only requires state x_K

⇒ **Efficient Inference**

Convolution in Training:

$$\begin{aligned}x_0 &= \bar{B}u_0 \\x_1 &= \bar{A}\bar{B}u_0 + \bar{B}u_1 \\x_2 &= \bar{A}^2\bar{B}u_0 + \bar{A}\bar{B}u_1 + \bar{B}u_2 \\&\vdots \\y_0 &= C\bar{B}u_0 \\y_1 &= C\bar{A}\bar{B}u_0 + C\bar{B}u_1 \\y_2 &= C\bar{A}^2\bar{B}u_0 + C\bar{A}\bar{B}u_1 + C\bar{B}u_2\end{aligned}\tag{4}$$

Convolution Kernel:

$$y = \bar{K} \cdot u\tag{5}$$

S4 vs. Mamba

S4 Model

- Fixed parameters: $\bar{A}$, $\bar{B}$, $\bar{C}$
- Time-invariant dynamics
- Efficient convolution-based training

Mamba Model

- Input-dependent: $\bar{A}(u)$, $\bar{B}(u)$, $\bar{C}(u)$
- Selective state space
- Hardware-aware scan for efficiency

Key Innovation: Mamba's selectivity enables better context-aware predictions, while maintaining computational efficiency

Source: [3, pp. 2–9]

Theoretical Analysis of SSM (compared to RNN, Transformer)

Comparison	RNNs	Transformers	SSMs
Training Speed	Slow (sequential)	Fast (parallelizable)	Fast (parallelizable)
Inference Speed	Fast (linear)	Slow (quadratic)	Fast (linear)
Complexity	$O(LD^2)$	$O(L^2D)$	$O(LD^2)$
Modeling Capabilities	State	Attention	State

Table: Comparison of sequence modeling approaches

Empirical Analysis: Overview

S4 Evaluation

Synthetic Benchmarks:

- Original: Long Range Arena
- Critique: Never Train from Scratch

Sequence Tasks:

- Original: Speech Commands
- Critique: Machine Translation

⇒ **Idea:** Compare benchmarks of SSM papers to comparable benchmarks of independent papers to evaluate objective performance

Mamba Evaluation

Long-Range Dependencies:

- Original: Induction Heads

Language Modeling:

- Original: Standard Language Modeling
- Independent: Machine Translation

Extended Sequences:

- Original: Very-long contexts

S4: Long Range Arena

Long Range Arena Benchmark by Gu et al.:

Table: LRA performance scores for Transformers, S4, and RNNs (linear and ReLU).
Symbols: x = failed, – = not tested.

Model	ListOps	Text	Retrieval	Image	Path	Path-X
Transformer	36.37	64.27	57.46	42.44	71.40	×
Reformer	37.27	56.10	53.40	38.07	68.50	×
Performer	18.01	65.40	53.82	42.77	77.05	×
Lin-Transf.	16.13	65.90	53.09	42.34	75.30	×
S4	59.60	86.82	90.90	88.65	94.20	96.35
Mamba	38.02	82.98	72.14	69.82	69.26	67.32
RNN-Lin	50.4	89.1	89.1	–	×	×
RNN-ReLU	37.6	88.0	88.5	–	×	×

⇒ **Key finding:** S4 outperforms all other models, including Mamba

Source: Adapted from [4, pp. 8,25f.], [6, p. 22f.], [10, p. 11f.]

S4: Long Range Arena

Critique of Long Range Arena by Amos et al.:

Table: LRA performance score comparison between self-pretrained (referred to as “Masked SPT” by authors) and scratch-trained models for Transformers and S4. Authors claim similarity in approach and comparability to S4 paper by Gu et al.

Model	ListOps	Text	Retrieval	Image	Path	Path-X
Transformer	36.37	64.27	57.46	42.44	71.40	×
Tr. + Masked SPT	59.75	89.27	88.64	74.22	88.45	87.73
S4	59.60	86.82	90.90	88.65	94.20	96.35
S4 + Masked SPT	61.25	90.34	88.74	89.36	94.92	96.94

⇒ **Key finding:** Pretraining significantly improves both architectures, S4 and Transformer perform comparably well

Source: Adapted from [1, p. 5]

Mamba: Language Modeling

Language Modeling Benchmark by Gu and Dao:

Table: Zero-Shot Evaluations performance results of Language Modeling tasks for Hybrid H3 (S4), Pythia (Transformer), and Mamba.

Model	Size	LAMBADA	PIQA	WinoGrande	Avg
Hybrid H3	360M	48.0	68.1	54.1	48.0
Pythia	410M	51.4	66.9	53.8	48.2
Mamba	370M	55.6	69.5	55.3	50.0
Hybrid H3	1.3B	49.6	71.3	56.9	53.0
Pythia	1.4B	61.7	71.0	57.2	55.2
Mamba	1.4B	64.9	74.2	61.5	59.7
Hybrid H3	2.7B	55.7	71.3	61.4	58.0
Pythia	2.8B	64.7	74.0	59.7	59.1
Mamba	2.8B	69.2	75.2	63.5	63.3

⇒ **Key Finding:** Mamba consistently outperforms comparable models across all size classes

Source: Adapted from [3, pp. 11f.,31f.]

Mamba: Machine Translation

Machine Translation Benchmark by Pitorro et al.:

Table: WMT23-CAT-10 BLEU scores for DE→EN and EN→DE. Multiple model configurations tested by authors (also Multi-Head-Attention (MHA and Encoder-Decoder)

Model	DE→EN	EN→DE
<i>Trained from Scratch</i>		
Transformer Enc-Dec	28.3	29.3
Transformer (best impl.)	29.8	29.1
Mamba	25.9	25.5
Mamba-MHA	27.8	25.9
Mamba Enc-Dec	31.4	30.1
<i>Finetuned</i>		
Mamba (430M)	25.6	22.5
Pythia-S (430M)	26.8	29.3
Mamba-M (1.4B)	32.5	27.5
Pythia-M (1.4B)	33.4	33.9

⇒ **Key Finding:** Pretrained Transformer Model outperforms Mamba for equal sizes

Source: Adapted from [7, p. 17]

Challenges in Empirical Analysis

Benchmark Methodology

- Studies use different configurations and training procedures
- Makes fair comparison difficult
- Example: Pretraining dramatically changes LRA results
- Very limited benchmarks that compare RNN, S4/Mamba, Transformers

Task-Specific Performance

- No model universally outperforms others
- Performance highly task-dependent, even within similar domains
- Example: Mamba excels at Language Modeling, but struggles in Machine Translation

Results of Empirical Analysis

S4:

- Performs well for synthetic benchmarks (comparable to Transformers, RNNs)
 - Fixed matrices limit model for more difficult tasks (Language Modeling)
- ⇒ Generally outperformed by Mamba model, that is considered as state-of-the-art SSM

Mamba:

- Comparable performance to Transformers in most benchmarks
- ⇒ Struggles fine-grained, token-level interactions (compared to Transformers)
- ⇒ Excels at extreme long sequences (compared to Transformers)
- ⇒ Mamba (SSMs) should be considered as an **alternative** for special applications
- ⇒ **Transformers remain state-of-the-art** for sequence-to-sequence problems

References I



Ido Amos, Jonathan Berant, and Ankit Gupta.

Never train from scratch: Fair comparison of long-sequence models requires data-driven priors, 2024.



Ian J. Goodfellow, Yoshua Bengio, and Aaron C. Courville.

Deep Learning.

Adaptive computation and machine learning. MIT Press, 2016.



Albert Gu and Tri Dao.

Mamba: Linear-time sequence modeling with selective state spaces, 2024.



Albert Gu, Karan Goel, and Christopher Ré.

Efficiently modeling long sequences with structured state spaces, 2022.



Christopher Olah.

Understanding lstm networks.

<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>, 2015.

Accessed: 2026-02-01.

References II

-  Antonio Orvieto, Samuel L Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De.
Resurrecting recurrent neural networks for long sequences, 2023.
-  Hugo Pitorro, Pavlo Vasylenko, Marcos Treviso, and André F. T. Martins.
How effective are state space models for machine translation?, 2024.
-  Haohao Qu, Liangbo Ning, Rui An, Wenqi Fan, Tyler Derr, Hui Liu, Xin Xu, and Qing Li.
A survey of mamba, 2025.
-  Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin.
Attention is all you need, 2023.
-  Annan Yu and N. Benjamin Erichson.
Block-biased mamba for long-range sequence processing, 2025.